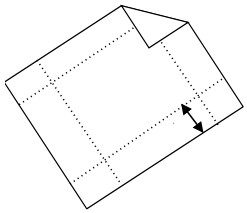
Integration

Chapter 12 · Original slide 70

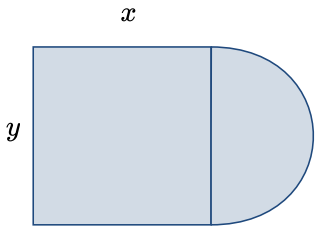
Optimisation Problems/Modelling

70



We have a sheet of A4 paper, which we want to fold into a cuboid. What height should we choose for the cuboid in order to maximise the volume?

These are examples of **optimisation problems**: we're trying to maximise/minimise some quantity by choosing an appropriate value of a variable that we can control.



We have 50m of fencing, and want to make a bear pen of the following shape, such that the area is maximised. What should we choose x and y to be?

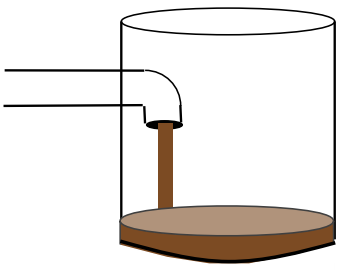
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'Rate of change'

71

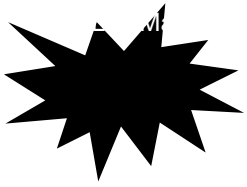
Up to now we've had y in terms of x, where $\frac{dy}{dx}$ means "the **rate** at which y changes with respect to x".

But we can use similar gradient function notation for other **physical quantities**.



A sewage container fills at a rate of 20 cm^3 per second.

How could we use appropriate notation to represent this?



Integration

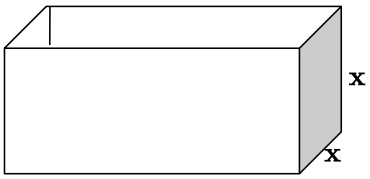
Chapter 12 · Original slide 72

Example Optimisation Problem

72

Optimisation problems in an exam usually follow the following pattern:

- There are 2 variables involved (you may have to introduce one yourself), typically lengths.
- There are expressions for **two different physical quantities**:
  - One is a **constraint**, e.g. "the surface area is 20cm^2".
  - The **other we wish to maximise/minimise**, e.g. "we wish to maximise the volume".
- We use the constraint to **eliminate one of the variables** in the latter equation, so that it is then **just in terms of one variable**, and we can then use differentiation to find the turning point.



[Textbook] A large tank in the shape of a cuboid is to be made from 54m^2 of sheet metal. The tank has a horizontal base and no top. The height of the tank is x metres. Two of the opposite vertical faces are squares.

a) Show that the volume, V m^3, of the tank is given by $V = 18x - \frac{2}{3}x^3$.

b) Given that x can vary, use differentiation to find the maximum or minimum value of V.

$\frac{dV}{dt}$

$V =$

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Test Your Understanding

73

Edexcel C2 May 2011 Q8

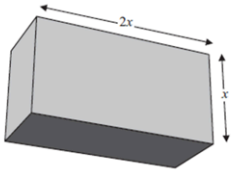


Figure 2

A cuboid has a rectangular cross-section where the length of the rectangle is equal to twice its width, x cm, as shown in Figure 2.

The volume of the cuboid is 81 cubic centimetres.

(a) Show that the total length, L cm, of the twelve edges of the cuboid is given by $L = 12x + \frac{162}{x^2}$.

\hfill (3)

(b) Use calculus to find the minimum value of L. \hfill (6)

(c) Justify, by further differentiation, that the value of L that you have found is a minimum. \hfill (2)

(b) $\frac{dL}{dx}$

Either 1

Correct $\{\frac{dL}{dx} =$

$L' = 0$

$x = \sqrt[3]{2}$

$\{x = 3$

Substitu

$54 \qua$

(c) {For

Correct f

$\frac{972}{x^3}$ an

(a)

$\{V =\}$

$\{L = 2($

$y = \frac{81}{2x^2}$

Making y

So, $L =$

Correct s

Integration

Chapter 12 · Original slide 74

Exercise 12K – Pearson Pure Y1 page 81

74

6 A rectangular garden is fenced on three sides, and the house forms the fourth side of the rectangle.

a Given that the total length of the fence is $80\,\text{m}$, show that the area, A, of the garden is given by the formula $A = y(80 - 2y)$, where y is the distance from the house to the end of the garden.

b Given that the area is a maximum for this length of fence, find the dimensions of the enclosed garden, and the area which is enclosed.

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Exercise 12K – Pearson Pure Y1 page 81

75

7 A closed cylinder has total surface area equal to 600π.
a Show that the volume, $V\,\text{cm}^3$, of this cylinder is given by the formula $V = 300\pi r - \pi r^3$, where $r\,\text{cm}$ is the radius of the cylinder.
b Find the maximum volume of such a cylinder.

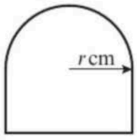
Integration

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Exercise 12K – Pearson Pure Y1 page 81

76

9 A shape consists of a rectangular base with a semicircular top, as shown.



a Given that the perimeter of the shape is $40\,\text{cm}$, show that its area, $A\,\text{cm}^2$, is given by the formula

$A = 40r - 2r^2 - \frac{\pi r^2}{2}$

where r cm is the radius of the semicircle. \hfill (2 marks)

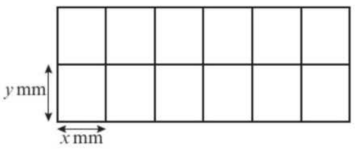
b Hence find the maximum value for the area of the shape. \hfill (4 marks)

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Exercise 12K – Pearson Pure Y1 page 81

77

10 The shape shown is a wire frame in the form of a large rectangle split by parallel lengths of wire into 12 smaller equal-sized rectangles.



a Given that the total length of wire used to complete the whole frame is $1512\,\text{mm}$, show that the area of the whole shape, $A\,\text{mm}^2$, is given by the formula

$A = 1296x - \frac{108x^2}{7}$

where x mm is the width of one of the smaller rectangles. \hfill (4 marks)

b Hence find the maximum area which can be enclosed in this way. \hfill (4 marks)

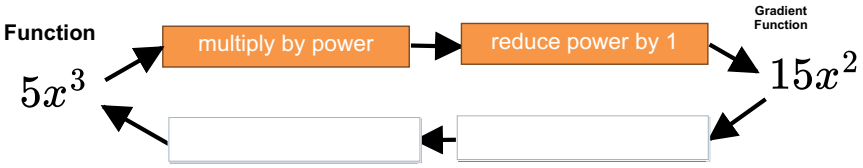
Integration

Chapter 13 · Original slide 3

Integrating x^n terms

3

Integration is the **opposite of differentiation**.
(For this reason it is also called '*antidifferentiation*')



However, there's one added complication…

Find y when $\frac{dy}{dx} = 3x^2$

Chapter 13 · Original slide 4

Examples

4

Find y when:

$\frac{dy}{dx} = 4x^3$ y c

$\frac{dy}{dx} = x^5$

$\frac{d}{d}$

Textbook Error on Pg289: "When integrating polynomials", but Example 3 is not a polynomial because it has negative and fractional powers.

Integration

Chapter 13 · Original slide 5

More Fractional Examples

5

Find y when:

$\frac{dy}{dx} = \square x^{-\frac{1}{2}}$ $\square 3x^{\frac{1}{2}} + c$

$\frac{dy}{dx} = 5x \square$

$\square$

$\frac{dy}{dx} \square$

Chapter 13 · Original slide 6

Test Your Understanding

6

Find $f(x)$ when:

$f'(x) = 2x + 7$ $\square$

$f'(x) = x^2 - 1$ $\square$

$\square = \frac{2}{x^7} = 2x^{-7}$ $\square$

$\square) = \sqrt[3]{x} = x^{\frac{1}{3}}$ $\square$

$f'(x) = 33x^{\frac{5}{6}}$ $\square$

Note: In case you're wondering what happens if $\frac{dy}{dx} = \frac{1}{x} = x^{-1}$, the problem is that after adding 1 to the power, we'd be dividing by 0. You will learn how to integrate $\frac{1}{x}$ in Year 2.

Integration

Chapter 13 · Original slide 7

Exercise 13A – Y1 pages 289-290

7

4. Find y given that $\frac{dy}{dx} = (2x+3)^2$

5. Find $f(x)$ given that
$f'(x) = 3x^{-2} + 6x^{\frac{1}{2}} + x - 4$

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Exercise 13A – Y1 pages 290

8

Challenge. Find y given that $\frac{dy}{dx} = \left(2\sqrt{x} - x^2\right)\left(\frac{3+x}{x^5}\right)$

Integration

Chapter 13 · Original slide 9

Integration notation

9

The following notation could be used to differentiate an expression:

$$\frac{d}{dx}\left(5x^2\right) = 10x$$

The dx here means differentiating "with respect to x".

There is similarly notation for integrating an expression:

$$\int 10x \ \ dx = 5x^2 + c$$

"Integrate…"

"…this expression"

"…with respect to x"
(the dx is needed just as it was needed in the differentiation notation at the top of this slide)

This is known as **indefinite integration**, in contrast to definite integration, which we'll see later in the chapter.

It is called 'indefinite' because the exact expression is unknown (due to the $+c$).

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Examples

10

Note: The brackets are required if there's multiple terms.

Find $\int (x^{-\frac{3}{2}} + 2)dx$

$= \quad + c$

Find $\int (6t^2 - 1)dt$

Note the dt instead of dx.

Find $\int (px^3 + q)dx$ where p and q are constants.

Integration

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Test Your Understanding

11

Edexcel C1 May 2014(R) Q4b

Given that $y = 2x^5 + \frac{6}{\sqrt{x}}$, $x > 0$, find in their simplest form

(b) $\int y \, \mathrm{d}x$ \hfill (3)

$y = 2x^5 +$
$\int y dx = \frac{}{}$

Chapter 13 · Original slide 12

Exercise 13B – Y1 pages 291-293

12

10 Given that $\mathrm{f}(x) = \frac{6}{x^2} + 4\sqrt{x} - 3x + 2$, $x > 0$, find $\int \mathrm{f}(x) \, \mathrm{d}x$.

11 Find $\int \left(8x^3 + 6x - \frac{3}{\sqrt{x}}\right) \mathrm{d}x$, giving each term in its simplest form.

Integration

Chapter 13 · Original slide 13

Exercise 13B – Y1 pages 291-293

13

12 a Show that $(2+5\sqrt{x})^2$ can be written as $4+k\sqrt{x}+25x$, where k is a constant to be found.

b Hence find $\int (2+5\sqrt{x})^2 \, \mathrm{d}x$.

13 Given that $y = 3x^5 - \frac{4}{\sqrt{x}}$, $x > 0$, find $\int y \, \mathrm{d}x$ in its simplest form.

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Exercise 13B – Y1 pages 291-293

14

14 $\int \left(\frac{p}{2x^2} + pq\right) \mathrm{d}x = \frac{2}{x} + 10x + c$

Find the value of p and the value of q.

15 $\mathrm{f}(x) = (2-x)^{10}$.

Given that x is small, and so terms in x^3 and higher powers of x can be ignored:

a find an approximation for $\mathrm{f}(x)$ in the form $A + Bx + Cx^2$

b find an approximation for $\int \mathrm{f}(x) \, \mathrm{d}x$.

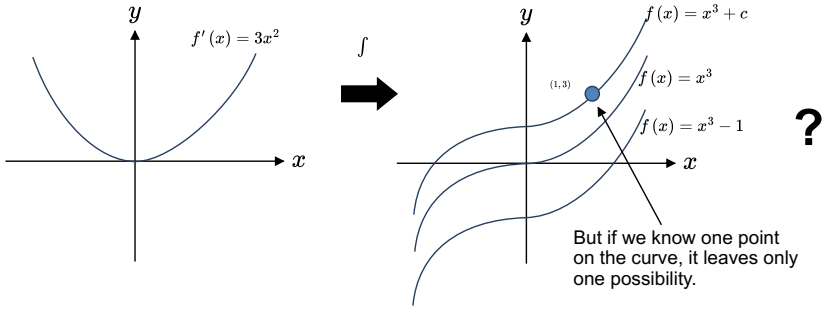
Integration

Chapter 13 · Original slide 15

Finding constant of integration

15

Recall that when we integrate, we get a constant of integration, which could be any real value. This means **we don't know what the exact original function was**.



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Test Your Understanding

16

Edexcel C1 May 2014 Q10

A curve with equation $y = \mathrm{f}(x)$ passes through the point $(4, 25)$.

Given that
$\mathrm{f}'(x) = \frac{3}{8}x^2 - 10x^{-1/2} + 1, \qquad x > 0$

(a) find $\mathrm{f}(x)$, simplifying each term. \hfill (5)

(b) Find an equation of the normal to the curve at the point $(4, 25)$.

Give your answer in the form $ax + by + c = 0$, where a, b and c are integers to be found.
\hfill (5)

Integration

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Exercise 13C – Y1 pages 294-295

17

4 The curve with equation $y = \mathrm{f}(x)$ passes through the point $(-1, 0)$. Given that $\mathrm{f}'(x) = 9x^2 + 4x - 3$, find $\mathrm{f}(x)$.

5 $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 3x^{-1/2} - 2x\sqrt{x}$, $x > 0$.

Given that $y = 10$ at $x = 4$, find y in terms of x, giving each term in its simplest form.

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Exercise 13C – Y1 pages 294-295

18

6 Given that $\dfrac{6x + 5x^{3/2}}{\sqrt{x}}$ can be written in the form $6x^p + 5x^q$,

a write down the value of p and the value of q.

Given that $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{6x + 5x^{3/2}}{\sqrt{x}}$ and that $y = 100$ when $x = 9$,

b find y in terms of x, simplifying the coefficient of each term.

Integration

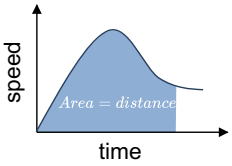
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Definite Integration

19

So far we've seen integration as '**the opposite of differentiation**', allowing us to find $y = f(x)$ when we know the gradient function $y = f'(x)$.

In practical settings however the most useful use of integration is that **it finds the area under a graph**. Remember at GCSE for example when you estimated the area under a speed-time graph, using trapeziums, to get the distance?
If you knew the equation of the curve, you could get the exact area!

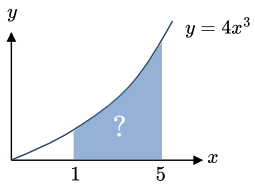


Before we do this, we need to understand how to find a **definite integral**:

These are known as **limits**, which give the values of x we're finding the area between.

We integrate as normal, but put expression in **square brackets**, meaning **we still need to evaluate the integrated expression using the limits**.

$$\int_1^5 4x^3 dx =$$



Write $(\dots) - (\dots)$ and evaluate the expression for each of the limits, top one first.

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Another Example

20

$$\int_{-3}^{3} x^2 + 1 dx = \left[\frac{1}{3}x\right.$$

$$= 12 - -12$$
$$= 24$$

You can use the $\left[\left[\int_b^a \square\right]\right]$ button on your calculator to evaluate definite integrals.
But only use it to check your answer.

Integration

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Problem Solving

21

Given that P is a constant and $\int_1^5 (2Px + 7)dx = 4P^2$, show that there are two possible values for P and find these values.

Remember: P is a constant, so just treat it as a number.

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Exercise 13D – Y1 page 297

22

4 Given that A is a constant and $\int_1^4 (6\sqrt{x} - A)\,dx = A^2$, show that there are two possible values for A and find these values.

Integration

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Exercise 13D – Y1 page 297

23

5 Use calculus to find the value of $\int_1^9 (2x - 3\sqrt{x})\,\mathrm{d}x$.

6 Evaluate $\int_4^{12} \frac{2}{\sqrt{x}}\,\mathrm{d}x$, giving your answer in the form $a + b\sqrt{3}$, where a and b are integers.

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Exercise 13D – Y1 page 297

24

7 Given that $\int_1^k \frac{1}{\sqrt{x}}\,\mathrm{d}x = 3$, calculate the value of k.

Integration

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EXTENSION

25

1 *[MAT 2009 1A]* The smallest value of

$$I(a) = \int_0^1 \left(x^2 - a\right)^2 dx$$

as a varies, is what?

$\mathbf{I(a) = \int_0^1 x^4 - 2ax^2 + a^2 dx}$

2 *[MAT 2015 1D]* Let $f(x) = \int_0^1 (xt)^2 dt$ and $g(x) = \int_0^x t^2 dt$

Let $A > 0$. Which of the following statements are true?

A) $g(f(A))$ is always bigger than $f(g(A))$

B) $f(g(A))$ is always bigger than $g(f(A))$

C) They are always equal.

D) $f(g(A))$ is bigger if $A < 1$, and $g(f(A))$ is bigger if $A > 1$.

E) $g(f(A))$ is bigger if $A < 1$, and $f(g(A))$ is bigger if $A > 1$.

(Official Sln) Evaluating: $\mathbf{f(x) = \frac{x^2}{3}}$ and $\mathbf{g(x) = \frac{x^3}{3}}$. Hence $\mathbf{g(f(A)) = \frac{A^6}{3^4}}$ and $\mathbf{f(g(A)) = \frac{A^6}{3^3}}$. Hence answer is (B).

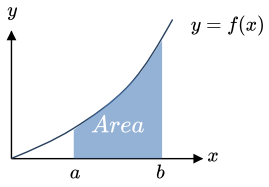
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Areas under curves

26

Earlier we saw that the definite integral $\int_a^b \mathbf{f(x)dx}$ gives the **area** between a positive curve $y = f(x)$, the **x-axis**, and the lines $x = a$ and $x = b$.

(We'll see why this works in a sec)



Find the area of the finite region between the curve with equation $y = 20 - x - x^2$ and the x-axis.

Integration

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Just for your interest…

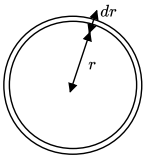
27

Why does integrating a function give you the area under the graph?

Part 1:

You're already familiar with the idea that gradient is the rate at which a quantity changes, and we consider an infinitesimally small change in the variable.
You could consider the gradient as the little bit you're adding on each time.

Here's some practical examples using formulae you covered in your younger years!

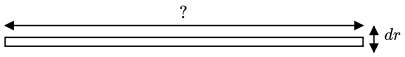


If you wanted to consider the rate at which the area of a circle increases with respect to the radius, consider a small change dr in the radius.
The change in area is an infinitely thin ring, which looks like you've drawn a circumference.
So what is this rate?
$A = \pi r^2$
$\therefore \frac{dA}{dr} = 2\pi r$



So to draw a shaded circle (which has area!), it's a bit like repeatedly drawing the circumference of a circle with gradually increasing radius.
Since the circumference is what the area is increasing by each time, the circumference is the gradient of the area of the circle, and conversely (by the definition of integration being the opposite of differentiation), the area is the integral of the circumference.

you can't add a length to an area
But by considering the infinitely thin width dr of the circumference you're drawing, it does have area!
$\frac{dA}{dr} = 2\pi r \quad \rightarrow \quad dA = 2\pi r dr$
i.e. the change in area, dA, is $2\pi r \times dr$
If we 'roll out' the added area we're adding each time, this forms a rectangle, whose length hopefully corresponds to the circumference of the circle:



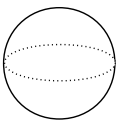
If the added area is $2\pi r dr$ and the thickness is dr, then the length is $\frac{2\pi r dr}{dr} = 2\pi r$ as expected.

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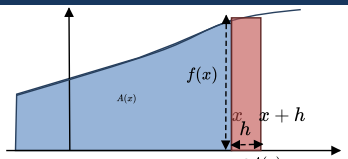
28

Part 2:

Consider the volume of a sphere:
$V = \frac{4}{3}\pi r^3$
$\frac{dV}{dr} = 4\pi r^2$



This works for a similar reason to before.



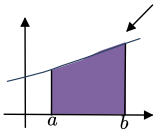
And the same principle applies to area under a graph. If $A(x)$ gives the area up to x, then what we're 'adding on each time' (i.e. the gradient) is sort of like drawing a line of length $f(x)$ at the right-most end each time (or more accurately, an infinitely thin rectangle of area $f(x) \times h$). Thus if $f(x)$ is the gradient of the area, then conversely the area is the integral of $f(x)$.

This gives us an **intuitive** sense of why it works, but we need a formal proof:

Using the diagram above, the area up to $x + h$, i.e. $A(x + h)$, is approximately the area up to x plus the added rectangular strip:
$$A(x + h) \approx A(x) + (f(x) \times h)$$
Rearranging:
$$f(x) \approx \frac{A(x+h) - A(x)}{h}$$
In the limit, as the rectangle becomes infinitely thin, this becomes an equality:
$$f(x) = \lim_{h \to 0} \frac{A(x+h) - A(x)}{h}$$
But we know from differentiating by first principles that:
$$A'(x) = \lim_{h \to 0} \frac{A(x+h) - A(x)}{h}$$
$$\therefore f(x) = A\prime(x)$$
And thus we have proven that the gradient of the area function is indeed $f(x)$ (and hence the integral of $f(x)$ the area).

One final Note: Why does $\int_b^a f(x)dx$ give the area between $x = a$ and $x = b$?

Since $f(x) = A\prime(x)$, the area function $A(x)$ is the integral of $f(x)$. Thus:
$$\int_b^a f(x)dx = [A(x)]_a^b \quad = A(b) - A(a)$$



The area between a and b is the area up to b, i.e. $A(b)$, with the area up to a, i.e. $A(a)$, cut out. This gives $A(b) - A(a)$ as expected. Because we obtained $A(x)$ by an integration, we have a constant of integration $+c$. But because of the subtraction $A(b) - A(a)$, these constants in each of $A(b)$ and $A(a)$ cancel, which explains why the constant is not present in definite integration.

This is known as the Fundamental Theorem of Calculus.

Integration

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Test Your Understanding

29

Edexcel C2 Jan 2013
Q9c

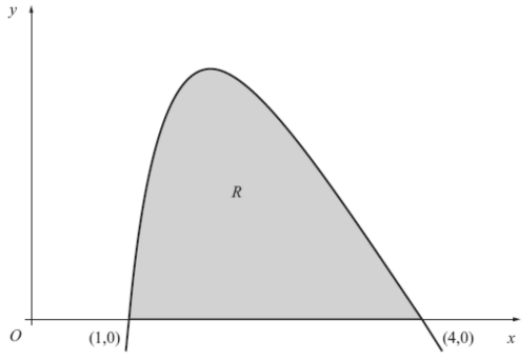


Figure 2

The finite region R, as shown in Figure 2, is bounded by the x-axis and the curve with equation
$y = 27 - 2x - 9\sqrt{x} - \frac{16}{x^2}, \qquad x > 0.$

The curve crosses the x-axis at the points $(1, 0)$ and $(4, 0)$.

(c) Use integration to find the exact value for the area of R. \hfill (6)

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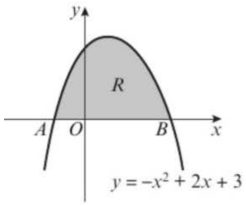
Exercise 13E – Y1 pages 299-300

30

8 The finite region R is bounded by the x-axis and the curve with equation $y = -x^2 + 2x + 3$, $x \geq 0$.
The curve meets the x-axis at points A and B.

a Find the coordinates of point A and point B.

b Find the area of the region R.



Integration

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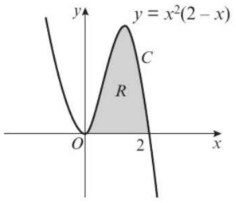
Exercise 13E – Y1 pages 299-300

31

9 The graph shows part of the curve C with equation $y = x^2(2 - x)$.

The region R, shown shaded, is bounded by C and the x-axis.

Use calculus to find the exact area of R. \hfill (5 marks)



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EXTENSION

32

[MAT 2007 1H] Given a function $f(x)$, you are told that

$\int_0^1 3f(x)dx + \int_1^2 2f(x)dx = 7$

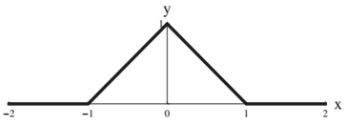
$\int_0^2 f(x)dx + \int_1^2 f(x)dx = 1$

It follows that $\int_0^2 f(x)dx$ equals what?

Because the area between $x=0$ and 2 is the sum of the area between 0 and 1, and between 1 and 2, it follows that

[MAT 2011 1G]

A graph of the function $y = f(x)$ is sketched on the axes below:



What is the value of $\int_{-1} f(x-1)dx$?

We first need to reflect on what part of the

Integration

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‘Negative Areas’

33

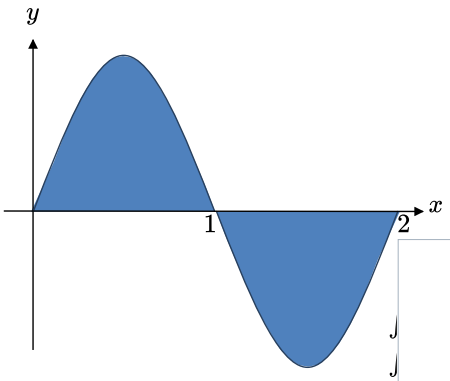
Sketch the curve $y = x(x-1)(x-2)$ (which expands to give $y = x^3 - 3x^2 + 2x$).
Now calculate $\int_0^2 x\,(x-1)\,(x-2)dx$. Why is this result surprising?

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Example

34

Find the total area bound between the curve $y = x(x-1)(x-2)$ and the x-axis.



Strategy:

Integration

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Test Your Understanding

35

Edexcel C2 May 2013 Q6

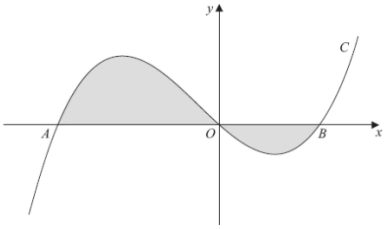


Figure 3 shows a sketch of part of the curve C with equation $y = x(x + 4)(x - 2)$.

The curve C crosses the x-axis at the origin O and at the points A and B.

(a) Write down the x-coordinates of the points A and B. \hfill (1)

The finite region, shown shaded in Figure 3, is bounded by the curve C and the x-axis.

(b) Use integration to find the total area of the finite region shown shaded in Figure 3. \hfill (7)

Figure 3

(a) S

(b) x
$\int(x^3$

or $\frac{x^4}{4}$
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or
$\left[\frac{x^4}{4}\right.$

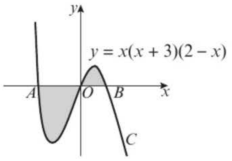
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Exercise 13F – Y1 pages 301-302

36

2 The graph shows a sketch of part of the curve C with equation $y = x(x + 3)(2 - x)$.



The curve C crosses the x-axis at the origin O and at points A and B.

a Write down the x-coordinates of A and B. \hfill (1 mark)

The finite region, shown shaded, is bounded by the curve C and the x-axis.

b Use integration to find the total area of the finite shaded region. \hfill (7 marks)

Integration

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Exercise 13F – Y1 pages 301-302

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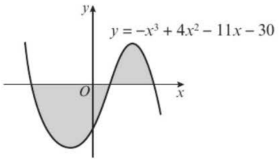
3 $f(x) = -x^3 + 4x^2 + 11x - 30$.

The graph shows a sketch of part of the curve with equation $y = -x^3 + 4x^2 + 11x - 30$.

a Use the factor theorem to show that $(x + 3)$ is a factor of $f(x)$.

b Write $f(x)$ in the form $(x + 3)(Ax^2 + Bx + C)$.

c Hence, factorise $f(x)$ completely.



d Hence, determine the x-coordinates where the curve intersects the x-axis.

e Hence, determine the total shaded area shown on the sketch.

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EXTENSION

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Extension

[MAT 2010 1I] For a positive number a, let

$$I(a) = \int_0^a \left(4 - 2^{x^2}\right) dx$$

Then $\frac{dI}{da} = 0$ when a is what value?

Hint: It's not actually even possible to integrate 2^{x^2}, but we can still sketch the graph. Reflect on what $\frac{dI}{da}$ actually means.

[STEP I 2014 Q3]

The numbers a and b, where $b > a \geq 0$, are such that

$$\int_a^b x^2 dx = \left(\int_a^b x dx\right)^2$$

(i) In the case $a = 0$ and $b > 0$, find the value of b.

(ii) In the case $a = 1$, show that b satisfies

$3b^3 - b^2 - 7b - 7 = 0$

Show further, with the help of a sketch, that there is only one (real) value of b that satisfies the equation and that it lies between 2 and 3.

(iii) Show that $3p^2 + q^2 = 3p^2q$, where $p = b + a$ and $q = b - a$, and express p^2 in terms of q. Deduce that $1 < b - a \leq \frac{4}{3}$

(iv)

Guidance for this problem on next slide.

Integration

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Guidance for Extension Problem 2

39

[STEP I 2014 Q3] The numbers a and b, where $b > a \geq 0$, are such that $\int_a^b x^2 dx = \left(\int_a^b x dx\right)^2$

(i) In the case $a = 0$ and $b > 0$, find the value of b.

(ii) In the case $a = 1$, show that b satisfies
$3b^3 - b^2 - 7b - 7 = 0$
Show further, with the help of a sketch, that there is only one (real) value of b that satisfies the equation and that it lies between 2 and 3.

(iii) Show that $3p^2 + q^2 = 3p^2q$, where $p = b + a$ and $q = b - a$, and express p^2 in terms of q.
Deduce that $1 < b - a \leq \frac{4}{3}$

This question was actually devised to address what happens when students misunderstand or mis-apply a "rule" of mathematics and *it turns out to give the right answer*. Part (i) starts you off gently: integrating both terms, squaring the RHS and solving very quickly gives $b = \frac{4}{3}$. Part (ii) develops in much the same way, but with a non-zero lower limit to the integrals, and we immediately see that the algebra gets much more involved. Importantly, it should be very clear that whatever expression materialises must have $(b - 1)$ as a factor (since setting $b = a$ would definitely give a zero area, thus trivially satisfying the given integral statement). This leads to the required cubic equation.

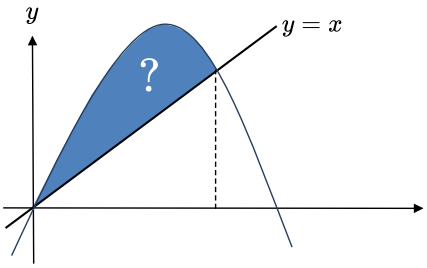
The final part of (ii) requires a mixture of different ideas (and can be done in a number of different ways). The most basic approach to demonstrating that a cubic curve has only one zero is to illustrate that both of its TPs lie on the same side of the x-axis (or to show there are no TPs). The popular *Change-of-Sign Rule* for continuous functions can be used to identify the position of this zero.

Having got you started with some simple lower limits, part (iii) develops matters more generally, and derives the (perhaps) surprising result that the exploration of this initial "stupid idea" requires b and a to be "not too far apart" to an extent that is easily identifiable.

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Areas between curves and lines

40



How could we find the area between the line and the curve?

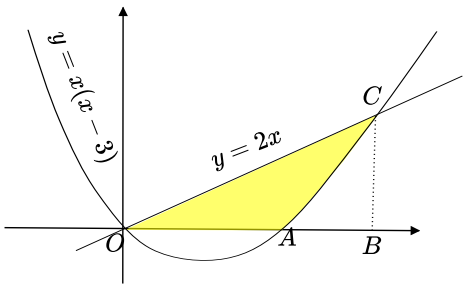
Determine the area between the lines with equations $y = x(4 - x)$ and $y = x$

Integration

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A Harder One

41



[Textbook] The diagram shows a sketch of the curve with equation $y = x(x-3)$ and the line with equation $y = 2x$.
Find the area of the shaded region OAC.

What areas should we subtract this time?
Start with triangle $\mathbf{OBC}$ and subtract

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Test Your Understanding

42

Edexcel C2 May 2012
Q5

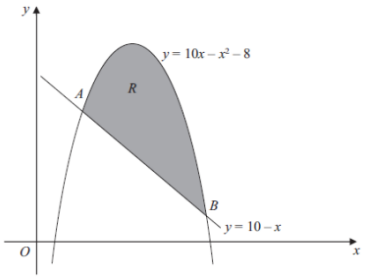


Figure 2 shows the line with equation $y = 10 - x$ and the curve with equation $y = 10x - x^2 - 8$.

The line and the curve intersect at the points A and B, and O is the origin.

(a) Calculate the coordinates of A and the coordinates of B. \hfill (5)

The shaded area R is bounded by the line and the curve, as shown in Figure 2.

(b) Calculate the exact area of R. \hfill (7)

a

b

Alternative Method:
If the top curve has equation $y = f(x)$ and the bottom curve $y = g(x)$, the area between them is:
$$\int_b^a (f(x) - g(x))dx$$
This means you can integrate a single expression to get the final area, without any adjustment required after.

Integration

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Exercise 13G – Y1 pages 304-306

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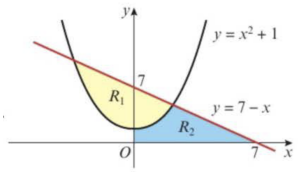
6 The diagram shows a sketch of part of the curve with equation $y = x^2 + 1$ and the line with equation $y = 7 - x$.

The finite region, R_1 is bounded by the line and the curve.

The finite region, R_2 is below the curve and the line and is bounded by the positive x- and y-axes as shown in the diagram.

a Find the area of R_1.

b Find the area of R_2.



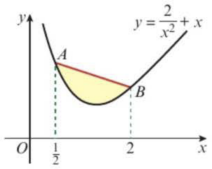
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Exercise 13G – Y1 pages 304-306

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8 The diagram shows part of a sketch of the curve with equation $y = \frac{2}{x^2} + x$. The points A and B have x-coordinates $\frac{1}{2}$ and 2 respectively.

Find the area of the finite region between AB and the curve.



Integration

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Exercise 13G – Y1 pages 304-306

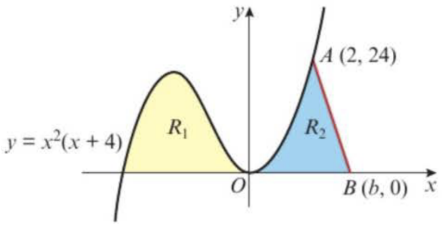
45

10 The sketch shows part of the curve with equation $y = x^2(x + 4)$. The finite region R_1 is bounded by the curve and the negative x-axis. The finite region R_2 is bounded by the curve, the positive x-axis and AB, where $A(2, 24)$ and $B(b, 0)$.

The area of R_1 = the area of R_2.

a Find the area of R_1.

b Find the value of b.



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EXTENSION

46

[MAT 2005 1A] What is the area of the region bounded by the curves $y = x^2$ and $y = x + 2$?

[MAT 2

$f(x) = a - x^2$

$g(x) = x^4 - a$

For precisely which values of $a > 0$ is the area of the region bounded by the x-axis and the curve $y = f(x)$ bigger than the area of the region bounded by the x-axis and the curve $y = g(x)$?

(Your answer should be an inequality in terms of a)